1. Pods em execução

```
=== Passo 1 : Pods (com IP e no) ===
> kubectl get pods -n mecanica -o wide

NAME                        READY   STATUS    RESTARTS   AGE     IP           NODE       NOMINATED NODE   READINESS GATES
backend-5c5cfb7758-hs4hn    1/1     Running   0          2m30s   10.244.0.25  minikube   <none>           <none>
backend-5c5cfb7758-qnzww    1/1     Running   0          20m     10.244.0.15  minikube   <none>           <none>
frontend-79fd677b8b-cw5rb   1/1     Running   0          20m     10.244.0.16  minikube   <none>           <none>
frontend-79fd677b8b-jfpxt   1/1     Running   0          20m     10.244.0.17  minikube   <none>           <none>
postgres-768fb75fd8-xxrz7   1/1     Running   0          88s     10.244.0.26  minikube   <none>           <none>
```

2. Services

```
> kubectl get services -n mecanica

NAME       TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
backend    ClusterIP   10.97.151.110    <none>        8080/TCP       21m
frontend   NodePort    10.107.49.40     <none>        80:30080/TCP   21m
postgres   ClusterIP   10.104.196.128   <none>        5432/TCP       21m
```

3. Deployments

```
> kubectl get deployments -n mecanica

NAME       READY   UP-TO-DATE   AVAILABLE   AGE
backend    2/2     2            2           22m
frontend   2/2     2            2           22m
postgres   1/1     1            1           22m
```

4. ConfigMaps, Secrets e volume persistente (PVC)

```
> kubectl get configmap,secret,pvc -n mecanica

NAME                          DATA   AGE
configmap/backend-config      4      22m
configmap/frontend-config     1      22m
configmap/kube-root-ca.crt    1      22m

NAME                     TYPE     DATA   AGE
secret/backend-secret    Opaque   3      22m
secret/postgres-secret   Opaque   3      22m

NAME                                  STATUS   VOLUME                                     CAPACITY   ACCESS MODES   STORAGECLASS   VOLUMEATTRIBUTESCLASS   AG
E
persistentvolumeclaim/postgres-pvc    Bound    pvc-87d013fb-63ef-4f8b-be24-ca03e7567460   1Gi        RWO            standard       <unset>                 22
m
```

5. Balanceamento de carga entre os Pods

```
> kubectl run lb-teste ... curl http://frontend/ (20x) e kubectl logs

X-Served-By: frontend-79fd677b8b-cw5rb
X-Served-By: frontend-79fd677b8b-cw5rb
X-Served-By: frontend-79fd677b8b-jfpxt
X-Served-By: frontend-79fd677b8b-jfpxt
X-Served-By: frontend-79fd677b8b-jfpxt
X-Served-By: frontend-79fd677b8b-jfpxt
X-Served-By: frontend-79fd677b8b-jfpxt
X-Served-By: frontend-79fd677b8b-jfpxt
X-Served-By: frontend-79fd677b8b-jfpxt
X-Served-By: frontend-79fd677b8b-jfpxt
X-Served-By: frontend-79fd677b8b-jfpxt
X-Served-By: frontend-79fd677b8b-cw5rb
X-Served-By: frontend-79fd677b8b-jfpxt
X-Served-By: frontend-79fd677b8b-cw5rb
X-Served-By: frontend-79fd677b8b-jfpxt
X-Served-By: frontend-79fd677b8b-cw5rb
X-Served-By: frontend-79fd677b8b-cw5rb
X-Served-By: frontend-79fd677b8b-jfpxt
X-Served-By: frontend-79fd677b8b-cw5rb
X-Served-By: frontend-79fd677b8b-jfpxt

--- Quantas respostas por Pod do frontend:

Count Name
----- ----
    7 X-Served-By: frontend-79fd677b8b-cw5rb
   13 X-Served-By: frontend-79fd677b8b-jfpxt
```

6. Auto-recuperação: Pod apagado e recriado

```
> kubectl delete pod <um backend> -n mecanica  e  kubectl get pods -l app=backend

ANTES:
NAME                       READY   STATUS    RESTARTS   AGE
backend-5c5cfb7758-hs4hn   1/1     Running   0          5m25s
backend-5c5cfb7758-qnzww   1/1     Running   0          23m

Apagando o Pod backend-5c5cfb7758-hs4hn ...
pod "backend-5c5cfb7758-hs4hn" deleted from mecanica namespace

5 SEGUNDOS DEPOIS (o antigo termina e um novo nasce):
NAME                       READY   STATUS    RESTARTS   AGE
backend-5c5cfb7758-lx2w2   0/1     Running   0          5s
backend-5c5cfb7758-qnzww   1/1     Running   0          23m
```

7. Backend de volta a 2 de 2 réplicas

```
> kubectl rollout status deploy/backend  e  kubectl get pods

deployment "backend" successfully rolled out

NAME                        READY   STATUS    RESTARTS   AGE
backend-5c5cfb7758-lx2w2    1/1     Running   0          26s
backend-5c5cfb7758-qnzww    1/1     Running   0          24m

NAME       READY   UP-TO-DATE   AVAILABLE   AGE
backend    2/2     2            2           24m
frontend   2/2     2            2           24m
postgres   1/1     1            1           24m
```

8. Persistência dos dados após apagar o Pod do banco

```
> kubectl delete pod -l app=postgres  e  SELECT na tabela mecanicos

ANTES (dados no banco):
 id |     nome     | especialidade
----+--------------+---------------
  1 | Carlos Teste | Motor
(1 row)

Apagando o Pod do Postgres ...
pod "postgres-768fb75fd8-xxrz7" deleted from mecanica namespace
Waiting for deployment "postgres" rollout to finish: 0 of 1 updated replicas are available...
deployment "postgres" successfully rolled out

Pod novo:
NAME                        READY   STATUS    RESTARTS   AGE
postgres-768fb75fd8-brfwp   1/1     Running   0          8s

DEPOIS (os dados continuam la):
 id |     nome     | especialidade
----+--------------+---------------
  1 | Carlos Teste | Motor
(1 row)
```

9. Aplicação funcionando no navegador

